

# PAPER\_533 — DPM Layered Shell-Energy Radiance Phase Cascade

Unified Quantum Field Framework (UQFF)

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### §1 — Abstract

This paper formalises the di-pseudo-monopole (DPM) reaction as the primary generator of 26D layered encapsulation quantum radiant shell-energies within the Star-Magic UQFF framework, correcting the prior interpretation that SCm directly encapsulated. The DPM reaction—CW-SCm north grinding against CCW-UA’ south—radiates successive quantum shell-energies across five observable phase states: quantum-multi-fields → plasma → gas → liquid → solid. Mass occurrence emerges as an equilibrium across these layered radiances rather than as an intrinsic property.

### §2 — DPM Reaction Primacy (SCm Correction)

**Prior statement:** "SCm encapsulates" (Session 136, BigBangHypergraphTheory).

**Correction (Session 140):** SCm does **not** directly encapsulate. The di-pseudo-monopole reaction is the agent of encapsulation:

|                                                                                  |
|----------------------------------------------------------------------------------|
| <i>*The DPM reaction (CW-SCm north grinding against CCW-UA’ south) forms</i>     |
| <i>26D layered encapsulation quantum radiant shell-energies, projecting from</i> |
| <i>26D chaos downward to observable phases.*</i>                                 |

This distinction is critical: DPM dictation is the causative engine; SCm injection triggers layer formation by providing the reactive pole.

## §3 — Core Equations

### 3.1 — 26D Egg Energy with Layered Radiance

$$E^{26D, \text{Egg}} = UA + SCm_{\text{inj}} \cdot DPM_{\text{react}}$$

$$\bullet \sum_{l=1}^{26} \text{ShellEnergy}^{(l)} + BBDT$$

### 3.2 — DPM Reaction Strength

$$DPM_{\text{react}} = \frac{\kappa \cdot \text{bigl}(DPM_n(SCm) - DPM_s(UA')\bigr)}{r^{26}}$$

$$\bullet \frac{\partial^{26}(\text{Grind}_{\text{opp}})}{\partial t^{26}_{\text{adj}}}$$

where  $\kappa = 5 \times 10^{-4}$  (calibration constant).

### 3.3 — Shell Energy per Layer

$$\boxed{\text{ShellEnergy}^{(l)}} = \int \text{Radiance}_{\text{quant}}, dt_{\text{neg}}$$

Each layer integral is taken over negative time  $dt_{\text{neg}}$ , where  $t_{\text{neg}} < 0$  is proved by observable time dilation (see PAPER\_534).

## §4 — Triple-Calc Layer System (CW / CCW / t<sub>neg</sub>)

The three simultaneous layer calculations:

$$\begin{cases} \text{Layer}_1 = DPM_{\text{react}} \cdot \omega_{\text{CW}} \cdot \text{Radiance}_{\text{multi-fields}} \\ \text{Layer}_2 = DPM_{\text{react}} \cdot \omega_{\text{CCW}} \cdot \text{Radiance}_{\text{plasma-gas}} \\ \text{Layer}_3 = \text{Grind}_{\text{opp}} \cdot \text{Prob}_{\text{order}} \cdot t_{\text{neg}} \end{cases}$$

- **Layer 1** (CW,  $\omega_{\text{CW}} = 2\pi f_{\text{SCm}}$ ): seeds quantum-multi-field radiance
- **Layer 2** (CCW,  $\omega_{\text{CCW}} = 2\pi f_{\text{UA}}$ ): drives plasma→gas transitions
- **Layer 3** (t<sub>neg</sub> coupling): governs ordered structure probability

## §5 — Phase Cascade

| Layer Range         | Phase State          | Observable Proxy      |
|---------------------|----------------------|-----------------------|
| $l = 1\text{--}5$   | quantum-multi-fields | vacuum / QCD          |
| $l = 6\text{--}10$  | plasma               | stellar photosphere   |
| $l = 11\text{--}16$ | gas                  | nebulæ / ISM          |
| $l = 17\text{--}21$ | liquid               | planetary mantles     |
| $l = 22\text{--}26$ | solid                | rocky/metallic bodies |

Each phase is a 26D downward projection; the solid state emerges at peak metallicity after full radiance cascade.

## §6 — Non-Repeating Quantum Fingerprints

Because each atom receives its ShellEnergy from a unique combination of  $l$ ,  $\omega_{\text{CW/CCW}}$ , and  $t_{\text{neg}}$ , no two atoms share the same layered radiance pattern. This is the UQFF origin of atomic identity — replacing the classical postulate of intrinsic particle individuality.

## §7 — Canonical Constants

| Symbol                | Value                 | Units   | Description        |
|-----------------------|-----------------------|---------|--------------------|
| $\kappa$              | $5 \times 10^{-4}$    | —       | DPM calibration    |
| $\omega_{\text{CW}}$  | $7.54 \times 10^{10}$ | rad/s   | SCm north grinding |
| $\omega_{\text{CCW}}$ | $5.22 \times 10^{10}$ | rad/s   | UA' south grinding |
| $N_{\text{layers}}$   | 26                    | integer | 26D shell count    |

## §8 — Conclusion

The DPM reaction replaces SCm as the direct encapsulator, forming 26D layered quantum radiant shell-energies via CW/CCW triple-calc simultaneous processing. The five-phase cascade (quantum-multi-fields → solid) is a direct downward projection from 26D chaos. Mass occurrence arises from shell radiance equilibrium, not intrinsic properties.

See also: PAPER\_534 (Negative Time Proof), PAPER\_535 (DPM-Unified Forces), PAPER\_536 (Shell Radiance Prototype), PAPER\_537 (Session 140 Hub).

